

MRPL Refinery — Thermal Performance Report:

Shell-and-Tube Heat Exchanger E-305

Document Control

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Reporting Period: August 2026

Equipment Specification

Parameter	Value
Equipment Tag	E-305
Equipment Type	Shell-and-Tube Heat Exchanger
Service	Crude Preheat - Hot Side: Atmospheric Residue
Design Duty	18.4 MW
Design U-value	420 W/m2K

Fouling Trend

Month	Measured U-value (W/m2K)	Fouling Factor (m2K/W)	Delta-T Approach (C)
May 2026	395	0.00015	18
June 2026	368	0.00033	22
July 2026	341	0.00051	27
August 2026	312	0.00071	33

Assessment

E-305 heat transfer coefficient has declined 25.7 percent from design over the reporting period, consistent with progressive fouling on the crude-side tubes. Approach temperature has widened from 18C to 33C, indicating reduced heat recovery and increased downstream furnace duty. Per MRPL Preventive Maintenance Manual Section 6.4, chemical cleaning or mechanical tube bundle cleaning is recommended when fouling factor exceeds 0.0006 m2K/W. E-305 has now exceeded this threshold and should be scheduled for cleaning at the next available turnaround window, or sooner if furnace fuel consumption continues to rise.

Recommended Action

- Schedule E-305 for mechanical tube bundle cleaning within 30 days.
- Monitor furnace fuel gas consumption weekly as an interim fouling indicator.
- Review crude slate composition for high-fouling-tendency feedstocks contributing to accelerated deposition.